Requirements Document

Item	Detail
Project	Roost, a verified student sublease marketplace for UC
Course	CS5001 Senior Design I, Assignment 9 (Week 9)
Team members	Danvanth Vanganuri, Jovan Jose Asker Fredy, Mohamed Aidja
Document version	2.0 (2026-10-28)
Companion document	Technical_Specification.md v2.0, which traces to this version
Prototype	roost-uc-sublease (commit `71d351b`, 2026-10-26); live at https://roost-uc-sublease.vercel.app

1. Version and Change Log

Version	Date	Change	Author
0.1	2026-09 (Week 4)	Baseline: US-01 to US-05, UC-01, UC-02, AC-01.1 to AC-02.2 (User_Stories.md v1.0).	Team
0.2	2026-10-14 (Week 7)	D2 Part 4 numbers NFR-01 to NFR-07; every story is Must.	Team
1.0	2026-10-28	Consolidated D0, D1, D2 and A8 into REQ-01 to REQ-16, NFR-01 to NFR-11 and CON-01 to CON-08. NFR-07 withdrawn (it became CON-01). Not submitted.	Jovan Jose Asker Fredy
2.0	2026-10-28	**Realigned to the Roost prototype and the advisor meetings** (Technical Specification §8). REQ-11 to REQ-14 withdrawn: property-manager approval was replaced by landlord consent (REQ-20). REQ-01 to REQ-04, REQ-06, REQ-07, REQ-10, REQ-15, REQ-16 and NFR-02, NFR-03, NFR-05, NFR-06, NFR-08, NFR-09, NFR-11 restated to match the built behavior. Added REQ-17 to REQ-29, NFR-12, CON-09. REQ-05 and NFR-04 (Fair Housing screening) are kept as requirements that are not built yet. User_Stories moves to v2.0.	Jovan Jose Asker Fredy
2.1	*pending*	Edits from the advisor review of these drafts.	*TBD*

ID rules. IDs are never renumbered or reused. A dropped requirement stays in its table marked WITHDRAWN, and the change is logged above. Story IDs (US, UC, AC) come from User_Stories.md v2.0.

2. Purpose and Scope

Purpose. This document states what Roost must do and for whom. Each requirement is a unit that the Week 10 test plan verifies. Each traces back to a user story and forward to a design element and a test case (Technical_Specification.md §6).

Product goal. Help UC students, faculty and staff fill or find term-length subleases safely. Only people with a UC email can use Roost. Listings are filtered by whole-term dates and distance from campus, and ranked by roommate fit. Subletting needs the landlord's consent, and contact details stay private until shared.

In scope for release 1.0 (class test in spring, demo Apr 16, 2027):

- The web app and the Expo phone app.
- Sign-in by an emailed link to a UC address.
- Posting listings with photos, landlord consent and the sublet-fee split.
- Swipe, list and map browsing, with date, distance, rent, bedroom and furnished filters.
- Roommate-fit ranking; nearest shuttle and bus stop; property-company Google rating.
- Saved listings and side-by-side compare.
- Real-time chat with opt-in contact sharing and 30-day deletion.
- Reporting a listing.
- Fair Housing screening of listings.

Out of scope: see the Won't list (§7.3).

3. Stakeholders and User Goals (Week 4, revised v2.0)

Category	Stakeholder	User goal	Stories
Primary	UC student leaving for a co-op (host)	Fill an empty room fast with someone trustworthy	US-01, US-08

Category	Stakeholder	User goal	Stories
Primary	Student relocating for one term (seeker)	Housing that covers the whole term, close to campus, with compatible roommates	US-02, US-06, US-07, US-10
Primary	UC faculty, staff, grad students (`@uc.edu`)	The same marketplace for short stays	US-01
Secondary	Property manager / landlord	Subletting only with their consent	US-08
Secondary	Parent or guarantor	The other party is a real UC person	US-01
Hidden (accessibility)	Screen-reader user	Search and contact without sighted help	US-04
Hidden (compliance)	Fair Housing reviewer	No discriminatory preference language published	US-05, US-09
Hidden (downstream)	UC Mail; Google Maps Platform; OpenStreetMap	Well-formed, rate-limited email; use within their terms	NFR-05, CON-08, CON-09

4. Glossary

Each term is defined once here and has this meaning everywhere in both documents.

Term	Definition
Host (subleaser)	A signed-in user who posts a listing. "Owner" in a requirement means the host of that listing.
Seeker (sublessee)	A signed-in user looking for a sublease, who starts a chat with a host.
UC email	An address ending exactly in `@mail.uc.edu` or `@uc.edu`.
Allowlist	The `email_allowlist` table of non-UC addresses that staff let in (advisor, testers). The API cannot read it.
Review account	`reviewer@roost-demo.test`, an allowlisted account that signs in with a password for the advisor. To be retired after review.
Sign-in link	A one-time link that Supabase Auth emails to the address. It is used for both sign-up and sign-in.
Signed-in user	A request carrying a valid Supabase session token (JWT). Anonymous requests are refused by row-level security.
Row-level security (RLS)	Postgres rules on every table that decide which rows each signed-in user may read or write.
Listing	One offer for a room or a whole unit. Its status is active, rented or removed. Sample listings (`is_sample`) are made-up demo data, labelled "Sample".
Landlord consent	The host's statement that the landlord allows subletting. A database check requires it on every listing.
Transfer type	Sublease, assignment or re-let.
Sublet fee / fee split	The landlord's fee for subletting, and the percentage (0 to 100) the host covers.
Whole-range match	A listing matches dates [from, to] when its start ≤ from and its end ≥ to. Overlap alone is not a match.
Distance	Straight-line miles from the UC campus center (39.1329, -84.5150) to the listing's coordinates.
Roommate preferences	Six 4-to-5-level scales (cleanliness, noise, hanging out at home, overnight guests, sleep schedule, approachability) plus smoke-free and cat/dog checkboxes. Users rate themselves; room listings carry the household's ratings.
Fit score	0 to 100% match between a seeker's preferences and a household. "Poor fit" is below 55%. Whole units rank as 75.
Shortlist	A user's saved listings. Compare shows 2 or 3 of them side by side.
Thread	A chat between one seeker and the host of one listing.
Contact sharing	Pressing Share in a thread, which lets the other participant read your email.
Report	A user's 3 to 200 character complaint about a listing. Staff act on reports.
Flagged phrase	Text that matches the Fair Housing phrase list. A listing with one is held for review (planned).
Freshness	Days since the host last confirmed "still available". After 10 days the listing is labelled "may be taken".
Cn, Gn, En, In	Component, glue module, external system and interface IDs from D0 to D2. TS §2 maps each to the prototype.
MoSCoW	Must (the release fails without it), Should (important, but the release works without it), Could (only if time remains), Won't (excluded from this release).

5. Functional Requirements

Verification: **T** = automated test, **D** = demonstration, **I** = inspection, **A** = analysis. "Built" = present in the prototype at commit `71d351b`.

ID	Requirement	Priority	Source story	Verification
REQ-01	The system shall create an account only for a UC email or an allowlisted address. It shall reject any other address before an email is sent, with "Only UC email addresses can join Roost".	**Must**	US-01; AC-01.2	T (rls-checks, auth API)
REQ-02	The system shall sign a user in only through a one-time emailed link (or the password of the review account). An expired or already used link shall create no session.	**Must**	US-01; AC-01.2	T
REQ-03	The system shall refuse every read or write of listings, profiles, chats, saved listings and reports from an anonymous request, and shall refuse the three API routes without a valid session (401).	**Must**	US-01	T
REQ-04	The system shall reject a listing whose end date is not after its start date, whose rent is outside $1 to $20,000 per month, whose title is outside 3 to 120 characters, that has more than 8 photos, or that lacks landlord consent. Nothing shall be written.	**Must**	AC-01.4	T
REQ-05	The system shall screen every listing's title and description against the versioned Fair Housing phrase list before it becomes active. A listing with at least one flagged phrase shall be held (not active), the host shown each flagged phrase, and the listing return 0 results for other users.	**Must**	US-05; AC-01.3	T
REQ-06	The system shall show a valid listing to other signed-in users within 60 s of a successful post.	**Must**	US-01; AC-01.1	T (timed)
REQ-07	The system shall show only active listings that are a whole-range match for the chosen dates, and shall offer Spring, Summer and Fall term presets.	**Must**	US-02; AC-03.1	T
REQ-08	The system shall let a signed-in seeker start one thread per listing with the host of an active listing that is not their own. It shall deliver 1 to 2,000 character messages to the other participant in real time, and only the two participants may read the thread.	**Must**	US-01; US-04	T
REQ-09	Only staff shall release or remove a held listing, and each decision shall be recorded with who and when.	**Should**	US-05	T
REQ-10	The system shall show the host their own listings, whatever the status, and let them re-confirm "still available" or mark a listing rented.	**Should**	US-01	T
REQ-11	WITHDRAWN (v2.0): owner proposes a sublessee for manager approval. Replaced by REQ-20.	—	(US-03, withdrawn)	—
REQ-12	WITHDRAWN (v2.0): manager approves or rejects. Replaced by REQ-20.	—	(US-03)	—
REQ-13	WITHDRAWN (v2.0): reminder after 5 business days.	—	(AC-02.2)	—
REQ-14	WITHDRAWN (v2.0): manager emailed when a listing is published.	—	(UC-01 step 10, v1)	—
REQ-15	The system shall show an unread-message count that updates in real time. There are no email alerts for messages this release.	**Could**	US-01	T
REQ-16	The system shall filter by a rent range chosen between $300 and $2,000, by bedrooms (1, 2, 3, 4+) and by furnished.	**Could**	US-02	T
REQ-17	The system shall compute each listing's distance from its geocoded address (it is never typed in) and filter by 1, 3, 5, 10 or 20 miles, or any distance.	**Should**	US-02; US-07	T
REQ-18	The system shall show each listing's nearest UC shuttle stop and Metro bus stop with their distance. Tapping a distance shall open a map of the stop, the listing and campus.	**Could**	US-07	T + D
REQ-19	The system shall collect roommate preferences after the first sign-in (editable later), rank listings by fit score, and hide poor fits until Show all, displaying how many are hidden.	**Should**	US-06; AC-03.2	T
REQ-20	The system shall require landlord consent on every listing (a database check). After consent it shall collect the sublet fee ($0 to $20,000) and the host's share (0 to 100%), and show the dollar split on the listing and in compare.	**Must**	US-08	T
REQ-21	The system shall reveal a participant's email to the other participant only after that participant presses Share. Each share shall post a system message that users cannot forge.	**Should**	US-11; AC-03.3	T
REQ-22	The system shall delete chat messages 30 days after they are sent (nightly, 03:17 UTC) and say so in the chat screen.	**Should**	US-11; AC-03.4	T
REQ-23	The system shall let a signed-in user report a listing once, with a 3 to 200 character reason, and let staff remove the listing (status "removed").	**Should**	US-09; AC-03.5	T + I
REQ-24	The system shall let a user save listings and compare 2 or 3 side by side, including rent, fees, distance and fit.	**Could**	US-10	T
REQ-25	The system shall accept up to 8 photos per listing, compress each to a 1,600 px long side plus a 480 px thumbnail before upload, and store them only in the host's own folder.	**Should**	US-08	T
REQ-26	The system shall present listings as a swipe deck (drag, buttons or arrow keys, with undo), as a list, and as a map, all sharing one set of filters.	**Could**	US-02	D
REQ-27	The system shall show the live Google star rating and review count of a listing's property company, fetched when the listing is viewed and never stored.	**Could**	US-02	T
REQ-28	The system shall label a listing "may be taken" when its host has not confirmed it as available for more than 10 days.	**Could**	US-02	T

ID	Requirement	Priority	Source story	Verification
REQ-29	The system shall offer suggested questions to ask a host, generated by a low-cost LLM with a daily spending cap.	**Could**	US-04; US-11	D

6. Non-Functional Requirements

ID	Type	Requirement	Priority	Source	Verification
NFR-01	Usability (accessibility)	Every route shall meet WCAG 2.1 AA with 0 serious or critical axe-core violations. Testers using NVDA and VoiceOver shall each complete search and message by keyboard only. Motion shall respect `prefers-reduced-motion`.	**Must**	US-04	T (axe-core) + D
NFR-02	Security (privacy)	No API response shall contain another user's email address unless that user shared it in the thread (REQ-21). A user shall read only their own profile, saved listings and reports.	**Must**	US-11; CON-02	T (rls-checks)
NFR-03	Security (access control)	Every table shall have RLS, and anonymous requests to every table, chat function and storage write shall be refused. All scripted RLS checks (28 at commit `71d351b`) shall pass on each release.	**Must**	US-01	T
NFR-04	Reliability (fail-closed)	100% of listings posted while screening is unavailable or errors shall be held or rejected, and never made active.	**Must**	US-05; CON-03	T (fault injection)
NFR-05	Security (abuse)	Address lookups shall be limited to 30 per user per minute, company searches to 20 and ratings to 60 (429 beyond). A host shall have at most 20 active listings. Sign-up and sign-in email shall stay within the Supabase Auth rate limit, with CAPTCHA before the class announcement.	**Should**	CON-08	T
NFR-06	Reliability (integrity)	Host names, chat names and listing titles shall be set by database triggers, never taken from the client. A failed insert shall leave no partial thread, message or listing row.	**Should**	US-01	T
NFR-07	WITHDRAWN (v1.0)	The budget requirement is a constraint; see CON-01.	—	—	—
NFR-08	Performance (load)	On a mid-range phone over 4G, Discover shall show its first card with Largest Contentful Paint ≤ 2.5 s at p75, and applying a filter shall update the results within 200 ms, with 200 active listings.	**Should**	US-02	T (Lighthouse mobile)
NFR-09	Performance (sign-in email)	95% of sign-in emails shall arrive in a `@mail.uc.edu` inbox (not junk) within 60 s of the request, through custom SMTP.	**Should**	US-01	T (inbox test)
NFR-10	Usability (task)	At least 4 of 5 first-time UC testers shall, without help, find a listing that covers given dates and message its host within 3 minutes.	**Should**	US-02; US-04	D (class test)
NFR-11	Reliability (availability)	The live site shall return 200 on ≥ 99% of 5-minute checks over the 7 days before the class test, on a Supabase plan that does not pause when idle.	**Should**	US-01	T (uptime monitor)
NFR-12	Security (secrets)	The shipped JavaScript shall contain no Supabase secret or service-role key and no Google Places key. Any browser Maps key shall be restricted to the site's address and the Maps JavaScript API.	**Should**	CON-09	I (bundle scan)

7. MoSCoW Summary

There are 36 active requirements: **13 Must (36%)**, 15 Should, 8 Could; 4 REQs and 1 NFR are withdrawn. The Must set is the smallest release that still serves every stakeholder group:

- Only UC people can get in.
- A host can post a valid, consented, screened listing.
- A seeker can find it by dates and message the host privately.
- The whole path works with a screen reader.

7.1 Must

REQ-01, REQ-02, REQ-03 (UC-only access) · REQ-04, REQ-05, REQ-06, REQ-20 (valid, screened, consented listings reach users) · REQ-07 (whole-range date match) · REQ-08 (private real-time chat) · NFR-01 (accessibility) · NFR-02, NFR-03 (privacy, access control) · NFR-04 (never publish unscreened).

REQ-05, NFR-04 and NFR-01 are Must but **not built yet**. They are the gap between the prototype and release 1.0 (Tasklist T4, T8, T17).

7.2 Should and Could

Should: REQ-09, REQ-10, REQ-17, REQ-19, REQ-21, REQ-22, REQ-23, REQ-25, NFR-05, NFR-06, NFR-08, NFR-09, NFR-10, NFR-11, NFR-12. **Could:** REQ-15, REQ-16, REQ-18, REQ-24, REQ-26, REQ-27, REQ-28, REQ-29.

Distance (REQ-17) and roommate fit (REQ-19) are Should, not Must: the advisor ranked them as first priorities, but a student can still find housing by dates without them. The swipe deck (REQ-26) is Could because the list view (REQ-07) already serves US-02.

7.3 Won't (this release)

These items are written down so they cannot creep back in. Reopening one needs a story and a change-log entry.

Won't	Why
Property-manager approval workflow and reminders (REQ-11 to REQ-14)	Team decision: landlord consent (REQ-20), requested by the advisor, covers it; the lease already gives the landlord control.
End-to-end encrypted chat	Advisor decision 2026-10-25: TLS in transit, AES-256 at rest, 30-day retention. E2EE would block review of reported chats and LLM suggestions.
UC single sign-on	Needs UC IT approval (CON-06); last on the checklist.
Payments, escrow, deposits, lease e-signature	Out of scope by design; legal and financial exposure.
Filters on protected characteristics	Fair Housing (CON-03).
Publishing to the App Store or Google Play	The phone app is tested through Expo Go only.
Live shuttle arrivals	Needs UC's TripShot data feed.
Google review text (only the rating and count)	Per-review attribution and higher cost (CON-09).
Self-service landlord or manager accounts	No story; landlords are not users this release.

8. Inherited Constraints

CON-01 to CON-05 come from the Week 3 constraint areas as D0 Part 6 records them. CON-06 to CON-08 come from the D0 decision log and the A8 plan. CON-09 is new in v2.0. TS §5 states what each one rules out or forces.

ID	Area	Constraint	Source
CON-01	Economic	$0 per month during development (free tiers). At launch, no more than Supabase Pro (about $25 per month). Google APIs capped by a daily quota and a budget alert.	Week 3; D0 Part 6; CHECKLIST (v2.0 restated; was about $7)
CON-02	Security / privacy	UC email addresses are not exposed to other users without consent; each user reaches only the rows the rules allow.	Week 3; D0 Part 6
CON-03	Legal	The Fair Housing Act (42 U.S.C. §3604(c)) and Ohio R.C. 4112 forbid discriminatory preference language in listings and in roommate wording.	Week 3; US-05; advisor prep 2026-10-21 Q6
CON-04	Social / ethical	Usable with assistive technology (WCAG 2.1 AA).	Week 3; US-04
CON-05	Hosting	Managed services only (Vercel, Supabase), with no servers for the team to patch.	Week 3; D0 Part 6
CON-06	Identity	UC SSO is not available without IT approval. Eligibility is proven by owning a UC mailbox.	D0 D-02; CHECKLIST
CON-07	Team / schedule	Three students, about 10 to 14 h per week, release by the Apr 16, 2027 expo.	A8 plan
CON-08	Downstream	UC Mail must receive authenticated email. Supabase's built-in sender only reaches the project team, so custom SMTP is required.	Week 4 hidden stakeholder; SUPABASE.md
CON-09	Third-party terms	Google Places: store only Place IDs, show Google attribution, no caching. OpenStreetMap Nominatim: at most 1 request per second, with an identifying User-Agent.	GOOGLE_REVIEWS.md (new v2.0)

9. Assumptions and Open Questions

Owners marked † are proposed; the team confirms them at the next meeting.

ID	Assumption or question	Owner	Due
A-01	Campus-scale demand: low thousands of users and a few hundred active listings per term. NFR-08 is sized on this.	Jovan	Recheck at M8 (2027-03-12)
A-02	Custom SMTP (a Gmail app password, about 500 emails per day) is enough for the class test until a domain exists.	Danvanth †	2026-11-13
A-03	Straight-line distance is acceptable for release 1.0.	Jovan	GQ-11
GQ-02	Phrase list v1 content, and the Ohio protected classes (ancestry, military status).	Jovan	M4 (2026-12-04)
GQ-03	(Closed v2.0.) Listings can now leave active: rented (host) or removed (staff).	—	—

ID	Assumption or question	Owner	Due
GQ-04	(Closed v2.0.) Manager approval withdrawn.	—	—
GQ-05	(Closed v2.0.) The email provider is Supabase Auth with custom SMTP; setup is still open (T19).	—	—
GQ-09	CON-01 to CON-05 come from D0's summary of the Week 3 essay; confirm against the essay. The budget change in CON-01 needs advisor agreement.	Jovan	2026-11-04
GQ-10	The TC-### IDs in the TS §6 matrix are reserved; the Week 10 test plan keeps them.	Mohamed	Week 10
GQ-11	Distance: straight-line (current) or driving/transit time? Maximum radius?	Advisor	Next meeting
GQ-12	Roommate categories and wording, the fairness check (CON-03), who sees a user's ratings, and whether hosts state preferences.	Advisor; Jovan	Next meeting
GQ-13	Who at UC (IT security, General Counsel) sets the chat-retention rule and the student-data policy?	Advisor	2026-11-13
GQ-14	Landlord consent: is a checkbox enough, or is proof needed (a lease clause upload)?	Advisor	Next meeting
GQ-15	LLM suggestions (REQ-29): what text may be sent to the provider, and what is the monthly cap?	Advisor	Before T27
GQ-16	Hide listings that are not re-confirmed after N days?	Team	2026-11-13
GQ-17	Metro (SORTA) GTFS stops to replace the sample bus stops.	Jovan †	2027-01-15
GQ-18	The default rent filter ($300 to $2,000) hides listings the database accepts (rent up to $20,000). Should the top of the slider mean "no limit"?	Mohamed †	2026-11-13